

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Hay Fever/ Rhinitis	Parker et al.	2009	Nationwide	3-17	PM _{2.5} , PM _{2.5-10} , NO ₂ , SO ₂	Annual (D8HourMax)	0.01655	0.00390	Logistic	Warm season; long term; 8-hour max from 24-hour mean using adjustment factor of 0.654, resulting in effective beta of 0.010818.

F.4.1 Gilliland et al. (2001)

Gilliland et al. (2001) examined the association between air pollution and school absenteeism among 4th grade school children (ages 9-10) in 12 southern Californian communities. The study was conducted from January through June 1996. The authors used school records to collect daily absence data and parental telephone interviews to identify causes. They defined illness-related absences as respiratory or non-respiratory. A respiratory illness was defined as an illness that included at least one of the following: runny nose/sneezing, sore throat, cough, earache, wheezing, or asthma attack. The authors used 15 and 30 day distributed lag models to quantify the association between ozone, PM₁₀, and NO₂ and incident school absences. Ozone levels were positively associated with all school absence measures and significantly associated with all illness-related school absences (non-respiratory illness, respiratory illness, URI and LRI). Neither PM₁₀ nor NO₂ was significantly associated with illness-related school absences, but PM₁₀ was associated with non-illness related absences. The health impact function for ozone is based on the results of the single pollutant model.

School Loss Days

Gilliland et al. (2001) defines an incident absence as an absence that followed attendance on the previous day and the incidence rate as the number of incident absences on a given day over the population at risk for an absence on a given day (i.e. those children who were not absent on the previous day). Since school absences due to air pollution may last longer than one day, an estimate of the average duration of school absences could be used to calculate the total avoided school loss days from an estimate of avoided new absences. A simple ratio of the total absence rate divided by the new absence rate would provide an estimate of the average duration of school absences, which could be applied to the estimate of avoided new absences as follows:

$$Duration = \frac{TotalAbsences}{NewAbsences}$$

$$\Delta TotalAbsences = -[incidence \times (e^{-\beta \times O_3} - 1)] \times duration \times pop$$

Since the function is log-linear, the baseline incidence rate (in this case, the rate of new absences) is multiplied by duration, which reduces to the total school absence rate. Therefore, the same result would be obtained by using a single estimate of the total school absence rate in the C-R function. Using this approach, we assume that the same

relationship observed between pollutant and new school absences in the study would be observed for total absences on a given day. As a result, the total school absence rate is used in the function below. The derivation of this rate is described in the section on baseline incidence rate estimation.

For all absences, the coefficient and standard error are based on a percent increase of 16.3 percent (95% CI -2.6 percent, 38.9 percent) associated with a 20 ppb increase in 8-hour average ozone concentration (2001, Table 6, p. 52).

A scaling factor is used to adjust for the number of school days in the ozone season. In the modeling program, the function is applied to every day in the ozone season (May 1 - September 30), however, in reality, school absences will be avoided only on school days. We assume that children are in school during weekdays for all of May, two weeks in June, one week in August, and all of September. This corresponds to approximately 2.75 months out of the 5 month season, resulting in an estimate of 39.3% of days ($2.75/5 \times 5/7$).

In addition, not all children are at-risk for a new school absence, as defined by the study. On average, 5.5% of school children are absent from school on a given day (U.S. Department of Education, 1996, Table 42-1). Only those who are in school on the previous day are at risk for a new absence ($1 - 0.055 = 94.5\%$). As a result, a factor of 94.5% is used in the function to estimate the population of school children at-risk for a new absence.

Incidence Rate: daily school absence rate = 0.055 (U.S. Department of Education, 1996, Table 42-1)

Population: population of children ages 9-10 not absent from school on a given day = 94.5% of children ages 9-10 (The proportion of children not absent from school on a given day (5.5%) is based on 1996 data from the U.S. Department of Education (1996, Table 42-1).)

Scaling Factor: Proportion of days that are school days in the ozone season = 0.393.

(Ozone is modeled for the 5 months from May 1 through September 30. We assume that children are in school during weekdays for all of May, 2 weeks in June, 1 week in August, and all of September. This corresponds to approximately 2.75 months out of the 5 month season, resulting in an estimate of 39.3% of days ($2.75/5 \times 5/7$).)

F.4.2 Ostro and Rothschild (1989)

Ostro and Rothschild (1989) estimated the impact of PM_{2.5} and ozone on the incidence of minor restricted activity days (MRADs) and respiratory-related restricted activity days (RRADs) in a national sample of the adult working population, ages 18 to 65, living in metropolitan areas. The study population is based on the Health Interview Survey (HIS), conducted by the National Center for Health Statistics. In publications from this ongoing survey, non-elderly adult populations are generally reported as ages 18-64. From the study, it is not clear if the age range stops at 65 or includes 65 year olds. We apply the C-R function to individuals ages 18-64 for consistency with other studies estimating impacts to non-elderly

adult populations. The annual national survey results used in this analysis were conducted in 1976-1981. Controlling for PM_{2.5}, two-week average ozone had a highly variable association with RRADs and MRADs. Controlling for ozone, two-week average PM_{2.5} was significantly linked to both health endpoints in most years. The C-R function for ozone is based on the co-pollutant model with PM_{2.5}.

The study is based on a “convenience” sample of non-elderly individuals. Applying the C-R function to this age group is likely a slight underestimate, as it seems likely that elderly are at least as susceptible to ozone as individuals under 65. A number of studies have found that hospital admissions for the elderly are related to ozone exposures (e.g., Schwartz, 1994b; Schwartz, 1995).

Minor Restricted Activity Days

The coefficient and standard error used in the C-R function are based on a weighted average of the coefficients in Ostro and Rothschild (1989, Table 4). The derivation of these estimates is described below.

Incidence Rate: daily incidence rate for minor restricted activity days (MRAD) = 0.02137 (Ostro and Rothschild, 1989, p. 243)

Population: adult population ages 18 to 64

The coefficient used in the C-R function is a weighted average of the coefficients in Ostro and Rothschild (1989, Table 4) using the inverse of the variance as the weight. The calculation of the MRAD coefficient and its standard error is exactly analogous to the calculation done for the work-loss days coefficient based on Ostro (1987).

$$\beta = \left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\sum_{i=1976}^{1981} \frac{1}{\sigma_{\beta_i}^2}} \right)$$

The standard error of the coefficient is calculated as follows, assuming that the estimated year-specific coefficients are independent:

$$\sigma_{\beta}^2 = \text{var} \left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\sum_{i=1976}^{1981} \frac{1}{\sigma_{\beta_i}^2}} \right) = \left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\gamma} \right) = \sum_{i=1976}^{1981} \text{var} \left(\frac{\beta_i}{\sigma_{\beta_i}^2 \times \gamma} \right)$$

This reduces down to:

$$\sigma_{\beta}^2 = \frac{1}{\gamma} \Rightarrow \sigma_{\beta} = \sqrt{\frac{1}{\gamma}} = 0.000658.$$

F.4.3 Parker et al. (2009)

Parker et al. (2009) investigated the associations between long-term O₃ exposure and respiratory allergies in an unrestricted population of children (aged 3-17 years) sampled from the United States National Health Interview Survey. Authors obtained symptom data from participant parents, who reported respiratory allergies on annual surveys. Parker et al. (2009) placed all study participants reporting symptoms of respiratory allergies or hay fever into a combined rhinitis group. Parker et al. (2009) linked annual averages of SO₂, NO₂, PM_{2.5}, and PM_{2.5-10} and warm season (May to September) O₃ averages to participant's addresses through ambient air pollution and meteorological data collected from US EPA Air Quality System monitors. The authors adjusted their logistic regression models for survey year, poverty-level, race/ethnicity, age, family structure, insurance coverage, usual source of care, education of adult, urban-rural status, region, and median county-level income.

Incidence, Hay Fever/Rhinitis

In a multi-pollutant model, the coefficient and standard error are based on the odds ratio of 1.18 (95% CI: 1.09-1.27) for a 10 ppb increase in 24-hour mean, warm season O₃ (Parker et al., 2009, Table 4).

The Health Impact Function was adjusted from the daily 24-hour mean metric to the daily 8-hour max metric using a ratio of 1/1.53 = 0.65359 (inverse of the ratio of 8-hour max to 24-hour mean ozone) (Anderson and Bell 2010, Table 2).

F.5 Asthma-Related Effects

Table F-5 summarizes the core health impacts functions used to estimate the relationship between ozone and asthma exacerbation. Below, we present a brief summary of each of the studies and any items that are unique to the study. Based on advice from the SAB-HES (U.S. EPA-SAB 2004), regardless of the age ranges included in the source epidemiology studies, we extend the applied population to ages 6 to 18, reflecting the common biological basis for the effect in children in the broader age group.

Table F-5. Core Health Impact Functions for Ozone and Asthma-Related Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Asthma Symptoms, Cough	Lewis et al.	2013	Detroit, MI	5-12		D8HourMax	0.00708	0.00372	Logistic	All year
Asthma Symptoms, Wheeze	Lewis et al.	2013	Detroit, MI	5-12		D8HourMax	0.00764	0.00410	Logistic	All year
Asthma Symptoms,	Lewis et al.	2013	Detroit, MI	5-12		D8HourMax	0.01140	0.00505	Logistic	All year